

applications. The DC operating point in a circuit with capacitors opened and inductors shorted is determined by Ngspice, which then specifies the analysis options on .OP, .DC and .TF control lines.

Similar to this analysis, Ngspice carries out the AC analysis for small signals, where the steady-state behaviour of the described system is determined at the specified set of stimulus frequencies with a single input node.

The transient analysis option of Ngspice is basically an extension of the time domain in DC analysis so it accepts the results of DC analysis to perform time-varying simulation. Time-independent DC sources are set to their DC values, whereas, in case of pole-zero analysis, Ngspice calculates the number of poles and zeros in the AC analysis of small signals.

Similarly, other analysis processes are performed at circuit temperature, which can be changed with TEMP and DTEMP options in Ngspice to allow temperature analysis at the same time.

Solves non-linear equations. Every circuit gives a linear or non-linear equation, of which the non-linear one is more difficult to solve during any simulation. Ngspice solves such equations using Newton Raphson algorithm, which is interactive and terminates on meeting required conditions. Although Ngspice is reliable in providing solutions to non-linear circuits, it fails during DC analysis if parameters are not set correctly or if connections are specified incorrectly.

Ngspice has been integrated into the most recent technology computer aided design (TCAD) tool, which is a general-purpose semiconductor simulator (gss) via a network interface.

Ngspice and its interfaces. Ngspice works with various graphical user interfaces (GUIs) depending upon the various operating systems and devices. The GUI on Windows is pretty simple and provides only program-text output, command history and command-line inputs when it comes to data inputs and uses Windows application programming interface (API) for plotting

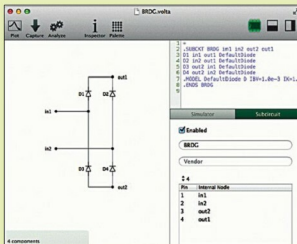


Fig. 4: Volta-ngspice GUI

data. Whereas on Linux, Ngspice uses a general console as a UI for input and X11 graphics system for output. CygWin interface that it works with is similar to Linux interface and also requires a console for input and X11 for output. However, it needs the following commands to initiate X11 Window Manager inside CygWin window before initiating Ngspice:

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$ export DISPLAY=:0.0
$ xwin -multiwindow -clipboard &
```

Ngspice also has a Web browser based interface in which simulation takes place on a remote server and another online version available for iPads, though neither is directly related to Ngspice development project. Even on iPads, simulation takes place on a remote server.

Can be compiled as a shared library. Ngspice, when added to an application, allows that application to gain control over the simulator because of its feature that lets it be compiled as a shared library. This is done by an interface offered by shared modules that not only export the functions that control the simulator but also call back those functions to get the feedback. The interface thus offered is not from Ngspice and is provided by the calling process where the user can develop and optimise elements within the interface without making even a slight change in Ngspice source code. The source code of the API thus shared is availa-

ble as sharedspice.c in a single C file, along with sharedspice.h as the corresponding header.

Returns data through call back functions. Ngspice uses call back functions to return data to the caller. These functions are globally-defined so that the C-coded Ngspice easily reaches to it. These functions can be easily identified as these are available as typedefs [for example, typedef

int (SendChar)(char, int, void);]. The address of these functions is obtained from ngSpice_Init function, when the caller is initialised with this command.

What the users say

Users have different views regarding Ngspice. A user at <http://sourceforge.net/> considers it to be an extremely useful tool for circuit designing, and approves Ngspice community to be a major benefit to this tool. He compares with tools like Hspice and Spectra, which are too expensive to consider. Whereas, some other users on the same website consider these to be perfect tools and appreciate their application.

Another user at www.electronicspoint.com/ considers Ngspice to be intelligent, providing values like 0.333333, 0.2 and 0.166666 because of its functionality to give errors when required, while he makes an error with PSpice providing numbers like 1/3, 1/5 and 1/6 that it considers to be 1.

Enhancements to Spice functionalities with the introduction of Ngspice is making this a popular tool with the available community support. The built-in models in this software act as a boon, allowing various types of analyses that are not always available in many other simulators.

Again, to get a free-of-cost simulator with so many features is quite difficult. ●

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